

ASEN ASEN OV

professor

基本信息

姓名: ASEN ASEN OV 所属院校: 格拉斯哥大学
邮箱: Asen.Asenov@glasgow.ac.uk 地区: United Kingdom

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

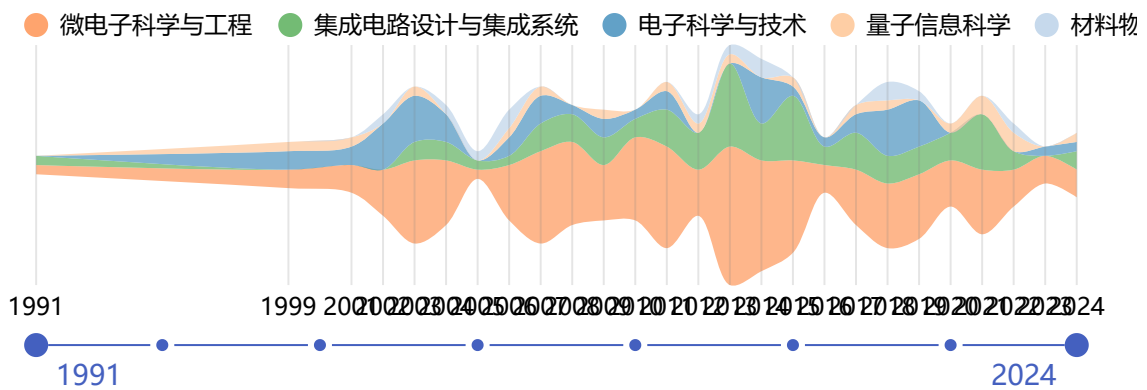
纳米电子与先进工艺

微电子器件与制造

个人简介

Professor ASEN ASEN OV holds the James Watt Chair in Electrical Engineering (Electronic & Nanoscale Engineering) at the University of Glasgow. His expertise lies in the field of Micro & Nanotechnology. Professor Asenov can be reached at Asen.Asenov@glasgow.ac.uk or via telephone at 01413304790. His office is located in R615 Level 6, Rankine Building, Glasgow G12 8LT. For further information, visit his ORCID profile at <https://orcid.org/0000-0002-9567-6366>.

近年发表的论文



Predictive Simulation of Nanosheet Transistors including the Impact of Access Resistance

2024 INTERNATIONAL CONFERENCE ON SIMULATION OF SEMICONDUCTOR PROCESSES AND DEVICES, SISPAD 2024

Dutta, Tapas; Adamu-Lema, Fikru; Xenid, Nikolas; Rezaei, Ali; Dixit, Ankit; Topaloglut, Ismail; Georgiev, Vihar; Asenov, Asen

被引用次数: 0

2024

Analysis of Random Discrete Dopants Embedded Nanowire Resonant Tunnelling Diodes for Generation of Physically Unclonable Functions

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Acharya, Pranav; Rezaei, Ali; Sengupta, Amretashis; Dutta, Tapas; Kumar, Naveen; Maciazek, Patryk; Asenov, Asen; Georgiev, Vihar

被引用次数: 0

2024

Unravelling the Impact of Random Dopant Fluctuations on Si-based 3nm NSFET: A NEGF Analysis

2024 IEEE 24TH INTERNATIONAL CONFERENCE ON NANOTECHNOLOGY, NANO 2024

Dixit, Ankit; Rezaei, Ali; Xeni, Nikolas; Kumar, Naveen; Dutta, Tapas; Topaloglu, Ismail; Aleksandrov, Preslav; Asenov, Asen; Georgiev, Vihar

被引用次数: 0

2024

The study of electron mobility on ultra-scaled silicon nanosheet FET

PHYSICA SCRIPTA

Liu, Tongfei; Rezaei, Ali; Yang, Kaige; Fan, Xuge; Acharya, Pranav; Georgiev, Vihar; Asenov, Asen; Ding, Jie

被引用次数: 1

2024

Single Event Transient Effects in Raised Source/Drain Double-Gate 1-T DRAM

8TH IEEE ELECTRON DEVICES TECHNOLOGY & MANUFACTURING CONFERENCE, EDTM 2024

Aneesh, Y. M.; Bindu, B.; Asenov, Asen

被引用次数: 0

2024

Hierarchical simulation of nanosheet field effect transistor: NESS flow

SOLID-STATE ELECTRONICS

Nagy, Daniel; Rezaei, Ali; Xeni, Nikolas; Dutta, Tapas; Adamu-Lema, Fikru; Topaloglu, Ismail; Georgiev, Vihar P.; Asenov, Asen

被引用次数: 3

2023

Convolutional Machine Learning Method for Accelerating Nonequilibrium Green's Function Simulations in Nanosheet Transistor

IEEE TRANSACTIONS ON ELECTRON DEVICES

Aleksandrov, Preslav; Rezaei, Ali; Dutta, Tapas; Xeni, Nikolas; Asenov, Asen; Georgiev, Vihar

被引用次数: 4

2023

Comprehensive mobility study of silicon nanowire transistors using multi-subband models

NANO EXPRESS

Medina-Bailon, Cristina; Nedjalkov, Mihail; Georgiev, Vihar; Selberherr, Siegfried; Asenov, Asen

被引用次数: 0

2023

How the Flat Field Transistor (FFT) can enable the continuation of DRAM scaling

2023 INTERNATIONAL CONFERENCE ON SIMULATION OF SEMICONDUCTOR PROCESSES AND DEVICES, SISPAD

Adamu-Lema, F.; Dutta, T.; Bensouiah, D.; Duan, M.; Asenov, A.

被引用次数: 0

2023

Electrolyte-Gated FET-based Sensing of Immobilized Amphoteric Molecules Including the Variability in Affinity of the Reactive Sites

2023 INTERNATIONAL CONFERENCE ON SIMULATION OF SEMICONDUCTOR PROCESSES AND DEVICES, SISPAD

Kumar, Naveen; Garcia, Cesar Pascual; Rezaei, Ali; Dixit, Ankit; Asenov, Asen; Georgiev, Vihar

被引用次数: 0

2023

STUDY OF INLAND WATERWAY TRANSPORT ROUTING OPTIONS UNDER CONDITIONS OF UNCERTAINTY IN NAVIGATIONAL RESTRICTIONS

TRANSPORT PROBLEMS

Pencheva, Velizara; Asenov, Asen; Georgiev, Ivan; Ivanov, Boril; Zanev, Ivelin

被引用次数: 0

2023

Carbon Nanotube SRAM in 5-nm Technology Node Design, Optimization, and Performance Evaluation--Part I: CNFET Transistor Optimization

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chen, Rongmei; Chen, Lin; Liang, Jie; Cheng, Yuanqing; Elloumi, Souhir; Lee, Jaehyun; Xu, Kangwei; Georgiev, Vihar P.; Ni, Kai; Debacker, Peter; Asenov, Asen; Todri-Sanial, Aida

被引用次数: 9

2022

Carbon Nanotube SRAM in 5-nm Technology Node Design, Optimization, and Performance Evaluation--Part II: CNT Interconnect Optimization

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chen, Rongmei; Chen, Lin; Liang, Jie; Cheng, Yuanqing; Elloumi, Souhir; Lee, Jaehyun; Xu, Kangwei; Georgiev, Vihar P.; Ni, Kai; Debacker, Peter; Asenov, Asen; Todri-Sanial, Aida

被引用次数: 1

2022

An Integral Methodology for Predicting Long-Term RTN

IEEE TRANSACTIONS ON ELECTRON DEVICES

Tok, Kean H.; Mehedi, Mehzabeen; Zhang, Jian F.; Brown, James; Ye, Zengliang; Ji, Zhigang; Zhang, Weidong; Marsland, John S.; Asenov, Asen; Georgiev, Vihar

被引用次数: 2

2022

Impact of the Figures of Merit (FoMs) Definitions on the Variability in Nanowire TFET: NEGF Simulation Study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Guan, Yunhe; Georgiev, Vihar P.; Asenov, Asen; Liang, Feng; Chen, Haifeng

被引用次数: 2

2022

Study of gate current in advanced MOS architectures

SOLID-STATE ELECTRONICS

Gauhar, Ghulam Ali; Chenchety, Abhishek; Yenugula, Hashish; Georgiev, Vihar; Asenov, Asen; Badami, Oves

被引用次数: 2

2022

Statistical device simulations of III-V nanowire resonant tunneling diodes as physical unclonable functions source

SOLID-STATE ELECTRONICS

Rezaei, Ali; Maciazek, Patryk; Sengupta, Amretashis; Dutta, Tapas; Medina-Bailon, Cristina; Asenov, Asen; Georgiev, Vihar P.

被引用次数: 1

2022

Effect of a Discontinuous Ag Layer on Optical and Electrical Properties of ZnO/Ag/ZnO Structures

COATINGS

Vitanov, Petko; Ivanova, Tatyana; Dikov, Hristosko; Terziyska, Penka; Ganchev, Maxim; Petkov, Nikolay; Georgiev, Yordan; Asenov, Asen

被引用次数: 5

2022

Improved compact model extraction of statistical variability in 5 nm nanosheet transistors and applied to SRAM simulations

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Li, Ruihan; Luo, Haowen; Wang, Yichen; Yuan, Zhengwu; Asenov, Asen; Miao, Xiangshui; Wang, Xingsheng

被引用次数: 1

2022

Quantum simulation investigation of work-function variation in nanowire tunnel FETs

NANOTECHNOLOGY

Guan, Yunhe; Carrillo-Nunez, Hamilton; Georgiev, Vihar P.; Asenov, Asen; Liang, Feng; Li, Zunchao; Chen, Haifeng

被引用次数: 4

2021

Stability and V_{min} analysis of ferroelectric negative capacitance FinFET based SRAM in the presence of variability

SOLID-STATE ELECTRONICS

Dutta, Tapas; Georgiev, Vihar; Asenov, Asen

被引用次数: 2

2021

Full-band quantum transport simulation in the presence of hole-phonon interactions using a mode-space approach

NANOTECHNOLOGY

Carrillo-Nunez, Hamilton; Medina-Bailon, Cristina; Georgiev, Vihar P.; Asenov, Asen

被引用次数: 3

2021

Simulation and Modeling of Novel Electronic Device Architectures with NESS (Nano-Electronic Simulation Software): A Modular Nano TCAD Simulation Framework

MICROMACHINES

Medina-Bailon, Cristina; Dutta, Tapas; Rezaei, Ali; Nagy, Daniel; Adamu-Lema, Fikru; Georgiev, Vihar P.; Asenov, Asen

被引用次数: 6

2021

Self-Consistent Enhanced S/D Tunneling Implementation in a 2D MS-EMC Nanodevice Simulator

MICROMACHINES

Medina-Bailon, Cristina; Padilla, Jose Luis; Sampedro, Carlos; Donetti, Luca; Gergiev, Vihar P.; Gamiz, Francisco; Asenov, Asen

被引用次数: 1

2021

KMC-based POM flash cell optimization and time-dependent performance investigation

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Ding, Jie; Mu, Dejiang; Badami, Oves; Medina-Bailon, Cristina; Chang, Xiaomin; Nagy, Daniel; Lapham, Paul; Georgiev, Vihar; Asenov, Asen

被引用次数: 0

2021

The analysis of static random access memory stability under the influence of statistical variability and bias temperature instability-induced ageing

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Ding, Jie; Asenov, Asen

被引用次数: 1

2021

The Extraction of Polyoxometalate Flash Compact Model and Corresponding Circuit Simulation

JOURNAL OF NANOELECTRONICS AND OPTOELECTRONICS

Ding, Jie; Yan, Guoliang; Adamu-Lema, Fikru; Dou, Yinke; Chen, Yan; Ding, Bing; Georgiev, Vihar P.; Asenov, Asen

被引用次数: 3

2021

Density Gradient Based Quantum-Corrected 3D Drift-Diffusion Simulator for Nanoscale MOSFETs

2021 IEEE 16TH NANOTECHNOLOGY MATERIALS AND DEVICES CONFERENCE (NMDC 2021)

Dutta, Tapas; Medina-Bailon, Cristina; Xen, Nikolas; Georgiev, Vihar P.; Asenov, Asen

被引用次数: 4

2021

Equivalent Circuit Macro-Compact Model of the 1T Bipolar SRAM Cell

2021 INTERNATIONAL CONFERENCE ON SIMULATION OF SEMICONDUCTOR PROCESSES AND DEVICES (SISPAD 2021)

Dutta, Tapas; Adamu-Lema, Fikru; Nagy, Daniel; Asenov, Asen; Nebesnyi, Valerii; Han, Jin-Woo; Widjaja, Yuniarto

被引用次数: 0

2021

PLANNING AND MODELING OF THE TIME FOR ACCEPTANCE AND STAY OF VEHICLES AT THE LOADING AND DISCHARGING POINTS

TRANSPORT PROBLEMS

Asenov, Asen; Pencheva, Velizara; Georgiev, Ivan

被引用次数: 1

2021

A Predictive 3-D Source/Drain Resistance Compact Model and the Impact on 7 nm and Scaled FinFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wu, Tao; Luo, Haowen; Wang, Xingsheng; Asenov, Asen; Miao, Xiangshui

被引用次数: 8

2020

Reliability-Aware Statistical BSIM Compact Model Parameter Generation Methodology

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ding, Jie; Asenov, Asen

被引用次数: 2

2020

A Kinetic Monte Carlo Study of Retention Time in a POM Molecule-Based Flash Memory

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Badami, Oves; Sadi, Toufik; Adamu-Lema, Fikru; Lapham, Paul; Mu, Dejiang; Georgiev, Vihar; Ding, Jie; Asenov, Asen

被引用次数: 4

2020

Quantum Enhancement of a S/D Tunneling Model in a 2D MS-EMC Nanodevice Simulator: NEGF Comparison and Impact of Effective Mass Variation

MICROMACHINES

Medina-Bailon, Cristina; Carrillo-Nunez, Hamilton; Lee, Jaehyun; Sampedro, Carlos; Padilla, Jose Luis; Donetti, Luca; Georgiev, Vihar; Gamiz, Francisco; Asenov, Asen

被引用次数: 7

2020

Nano-electronic Simulation Software (NESS): a flexible nano-device simulation platform

JOURNAL OF COMPUTATIONAL ELECTRONICS

Berrada, Salim; Carrillo-Nunez, Hamilton; Lee, Jaehyun; Medina-Bailon, Cristina; Dutta, Tapas; Badami, Oves; Adamu-Lema, Fikru; Thirunavukkarasu, Vasanthan; Georgiev, Vihar; Asenov, Asen

被引用次数: 22

2020

TCAD simulations and accurate extraction of reliability-aware statistical compact models

JOURNAL OF COMPUTATIONAL ELECTRONICS

Ding, Jie; Asenov, Asen

Thorough Understanding of Retention Time of Z2FET Memory Operation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Duan, M.; Navarro, C.; Cheng, B.; Adamu-Lema, F.; Wang, X.; Georgiev, V. P.; Gamiz, F.; Millar, C.; Asenov, A.

被引用次数: 12

2019

An Accurate Analytical Model for Tunnel FET Output Characteristics

IEEE ELECTRON DEVICE LETTERS

Guan, Yunhe; Li, Zunchao; Carrillo-Nunez, Hamilton; Zhang, Yefei; Georgiev, Vihar P.; Asenov, Asen; Liang, Feng

被引用次数: 9

2019

Simulation of the Impact of Ionized Impurity Scattering on the Total Mobility in Si Nanowire Transistors

MATERIALS

Sadi, Toufik; Medina-Bailon, Cristina; Nedjalkov, Mihail; Lee, Jaehyun; Badami, Oves; Berrada, Salim; Carrillo-Nunez, Hamilton; Georgiev, Vihar; Selberherr, Siegfried; Asenov, Asen

被引用次数: 23

2019

Mobility of Circular and Elliptical Si Nanowire Transistors Using a Multi-Subband 1D Formalism

IEEE ELECTRON DEVICE LETTERS

Medina-Bailon, Cristina; Sadi, Toufik; Nedjalkov, Mihail; Carrillo-Nunez, Hamilton; Lee, Jaehyun; Badami, Oves; Georgiev, Vihar; Selberherr, Siegfried; Asenov, Asen

被引用次数: 16

2019

Machine Learning Approach for Predicting the Effect of Statistical Variability in Si Junctionless Nanowire Transistors

IEEE ELECTRON DEVICE LETTERS

Carrillo-Nunez, Hamilton; Dimitrova, Nadezhda; Asenov, Asen; Georgiev, Vihar

被引用次数: 75

2019

Investigation of Pt-Salt-Doped-Standalone-Multiwall Carbon Nanotubes for On-Chip Interconnect Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Liang, Jie; Chen, Rongmei; Ramos, Raphael; Lee, Jaehyun; Okuno, Hanako; Kalita, Dipankar; Georgiev, Vihar; Berrada, Salim; Sadi, Toufik; Uhlig, Benjamin; Lilienthal, Katherina; Dhavamani, Abitha; Konemann, Fabian; Gotsmann, Bernd; Goncalves, Goncalves; Chen, Bingan; Asenov, Asen; Dijon, Jean; Todri-Sanial, Aida

被引用次数: 15

2019

Multisubband Ensemble Monte Carlo Analysis of Tunneling Leakage Mechanisms in Ultrascaled FDSOI, DGSOI, and FinFET Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Medina-Bailon, Cristina; Padilla, Jose L.; Sadi, Toufik; Sampedro, Carlos; Godoy, Andres; Donetti, Luca; Georgiev, Vihar P.; Gamiz, Francisco; Asenov, Asen

Trigger-When-Charged: A Technique for Directly Measuring RTN and BTI-Induced Threshold Voltage Fluctuation Under Use-Vdd

IEEE TRANSACTIONS ON ELECTRON DEVICES

Manut, Azrif; Gao, Rui; Zhang, Jian Fu; Ji, Zhigang; Mehedi, Mehzaheen; Zhang, Wei Dong; Vigar, David; Asenov, Asen; Kaczer, Ben

被引用次数: 9

2019

Comprehensive Study of Cross-Section Dependent Effective Masses for Silicon Based Gate-All-Around Transistors

APPLIED SCIENCES-BASEL

Badami, Oves; Medina-Bailon, Cristina; Berrada, Salim; Carrillo-Nunez, Hamilton; Lee, Jaeyhun; Georgiev, Vihar; Asenov, Asen

被引用次数: 15

2019

Variability Study of MWCNT Local Interconnects Considering Defects and Contact Resistances-Part I: Pristine MWCNT

IEEE TRANSACTIONS ON ELECTRON DEVICES

Chen, Rongmei; Liang, Jie; Lee, Jaehyun; Georgiev, Vihar P.; Ramos, Raphael; Okuno, Hanako; Kalita, Dipankar; Cheng, Yuanqing; Zhang, Liuyang; Pandey, Reetu R.; Amoroso, Salvatore; Millar, Campbell; Asenov, Asen; Dijon, Jean; Todri-Sanial, Aida

被引用次数: 9

2018

Variability Study of MWCNT Local Interconnects Considering Defects and Contact Resistances-Part II: Impact of Charge Transfer Doping

IEEE TRANSACTIONS ON ELECTRON DEVICES

Chen, Rongmei; Liang, Jie; Lee, Jaehyun; Georgiev, Vihar P.; Ramos, Raphael; Okuno, Hanako; Kalita, Dipankar; Cheng, Yuanqing; Zhang, Liuyang; Pandey, Reetu R.; Amoroso, Salvatore; Millar, Campbell; Asenov, Asen; Dijon, Jean; Todri-Sanial, Aida

被引用次数: 8

2018

Stochastic analysis of surface roughness models in quantum wires

COMPUTER PHYSICS COMMUNICATIONS

Nedjalkov, Mihail; Ellinghaus, Paul; Weinbub, Josef; Sadi, Toufik; Asenov, Asen; Dimov, Ivan; Selberherr, Siegfried

被引用次数: 8

2018

A review of the Z2-FET 1T-DRAM memory: Operation mechanisms and key parameters

SOLID-STATE ELECTRONICS

Cristoloveanu, S.; Lee, K. H.; Parihar, M. S.; El Dirani, H.; Lacord, J.; Martinie, S.; Le Royer, C.; Barbe, J. -Ch.; Mescot, X.; Fonteneau, P.; Galy, Ph.; Gamiz, F.; Navarro, C.; Cheng, B.; Duan, M.; Adamu-Lema, F.; Asenov, A.; Taur, Y.; Xu, Y.; Kim, Y-T.; Wan, J.; Bawedin, M.

被引用次数: 40

2018

Investigation of resistance switching in SiO_x RRAM cells using a 3D multi-scale kinetic Monte Carlo simulator

JOURNAL OF PHYSICS-CONDENSED MATTER

Sadi, Toufik; Mehonic, Adnan; Montesi, Luca; Buckwell, Mark; Kenyon, Anthony; Asenov, Asen

被引用次数: 27

2018

Process Variability for Devices at and beyond the 7 nm Node

ECS JOURNAL OF SOLID STATE SCIENCE AND TECHNOLOGY

Lorenz, J. K.; Asenov, A.; Baer, E.; Barraud, S.; Kluepfel, F.; Millar, C.; Nedjalkov, M.

被引用次数: 10

2018

Random Dopant-Induced Variability in Si-InAs Nanowire Tunnel FETs: A Quantum Transport Simulation Study

IEEE ELECTRON DEVICE LETTERS

Carrillo-Nunez, Hamilton; Lee, Jaehyun; Berrada, Salim; Medina-Bailon, Cristina; Adamu-Lema, Fikru; Luisier, Mathieu; Asenov, Asen; Georgiev, Vihar P.

被引用次数: 11

2018

Understanding Electromigration in Cu-CNT Composite Interconnects: A Multiscale Electrothermal Simulation Study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Jaehyun; Berrada, Salim; Adamu-Lema, Fikru; Nagy, Nicole; Georgiev, Vihar P.; Sadi, Toufik; Liang, Jie; Ramos, Raphael; Carrillo-Nunez, Hamilton; Kalita, Dipankar; Lilienthal, Katharina; Wislicenus, Marcus; Pandey, Reeturaj; Chen, Bingan; Teo, Kenneth B. K.; Goncalves, Goncalo; Okuno, Hanako; Uhlig, Benjamin; Todri-Sanial, Aida; Dijon, Jean; Asenov, Asen

被引用次数: 11

2018

Impact of Randomly Distributed Dopants on Ω -Gate Junctionless Silicon Nanowire Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Carrillo-Nunez, Hamilton; Mirza, Muhamad M.; Paul, Douglas J.; MacLaren, Donald A.; Asenov, Asen; Georgiev, Vihar P.

被引用次数: 8

2018

Variability Predictions for the Next Technology Generations of n-type SixGe_{1-x} Nanowire MOSFETs

MICROMACHINES

Lee, Jaehyun; Badami, Oves; Carrillo-Nunez, Hamilton; Berrada, Salim; Medina-Bailon, Cristina; Dutta, Tapas; Adamu-Lema, Fikru; Georgiev, Vihar P.; Asenov, Asen

被引用次数: 7

2018

Atomistic- to Circuit-Level Modeling of Doped SWCNT for On-Chip Interconnects

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Liang, Jie; Lee, Jaehyun; Berrada, Salim; Georgiev, Vihar P.; Pandey, Reeturaj; Chen, Rongmei; Asenov, Asen; Todri-Sanial, Aida

被引用次数: 7

2018

Impact of quantum confinement on transport and the electrostatic driven performance of silicon nanowire transistors at the scaling limit

SOLID-STATE ELECTRONICS

Al-Ameri, Talib; Georgiev, Vihar P.; Sadi, Toufik; Wang, Yijiao; Adamu-Lema, Fikru; Wang, Xingsheng; Amoroso, Salvatore M.; Towie, Ewan; Brown, Andrew; Asenov, Asen

被引用次数: 10

2017

A Survey of Carbon Nanotube Interconnects for Energy Efficient Integrated Circuits

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Todri-Sanial, Aida; Ramos, Raphael; Okuno, Hanako; Dijon, Jean; Dhavamani, Abitha; Wislicenus, Marcus; Lilienthal, Katharina; Uhlig, Benjamin; Sadi, Toufik; Georgiev, Vihar P.; Asenov, Asen; Amoroso, Salvatore M.; Brown, Andrew R.; Pender, Andrew; Millar, Campbell; Motzfeld, Fabian; Gotsmann, Bernd; Liang, Jie; Goncalves, Goncalo; Rupesinghe, Nalin; Teo, Ken

被引用次数: 21

2017

Z2-FET as Capacitor-Less eDRAM Cell For High-Density Integration

IEEE TRANSACTIONS ON ELECTRON DEVICES

Navarro, Carlos; Duan, Meng; Parihar, Mukta Singh; Adamu-Lema, Fikru; Coseman, Stefan; Lacord, Joris; Lee, Kyunghwa; Sampedro, Carlos; Cheng, Binjie; El Dirani, Hassan; Barbe, Jean-Charles; Fonteneau, Pascal; Kim, Seong-Il; Cristoloveanu, Sorin; Bawedin, Maryline; Millar, Campbell; Galy, Philippe; Le Royer, Cyrille; Karg, Siegfried; Riel, Heike; Wells, Paul; Kim, Yong-Tae; Asenov, Asen; Gamiz, Francisco

被引用次数: 27

2017

Extended Analysis of the Z2-FET: Operation as Capacitorless eDRAM

IEEE TRANSACTIONS ON ELECTRON DEVICES

Navarro, Carlos; Lacord, Joris; Parihar, Mukta Singh; Adamu-Lema, Fikru; Duan, Meng; Rodriguez, Noel; Cheng, Binjie; El Dirani, Hassan; Barbe, Jean-Charles; Fonteneau, Pascal; Bawedin, Maryline; Millar, Campbell; Galy, Philippe; Le Royer, Cyrille; Karg, Siegfried; Wells, Paul; Kim, Yong-Tae; Asenov, Asen; Cristoloveanu, Sorin; Gamiz, Francisco

被引用次数: 32

2017

Key Issues and Solutions for Characterizing Hot Carrier Aging of Nanometer Scale nMOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Duan, Meng; Zhang, Jian Fu; Ji, Zhigang; Zhang, Wei Dong; Ben Kaczer; Asenov, Asen

被引用次数: 22

2017

Simulation Study of Vertically Stacked Lateral Si Nanowires Transistors for 5-nm CMOS Applications

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Al-Ameri, Talib; Georgiev, Vihar P.; Adamu-Lema, Fikru; Asenov, Asen

被引用次数: 20

2017

Experimental and Simulation Study of Silicon Nanowire Transistors Using Heavily Doped Channels

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Georgiev, Vihar P.; Mirza, Muhammad M.; Dochioiu, Alexandru-Iustin; Adamu-Lema, Fikru; Amoroso, Salvatore M.; Towie, Ewan; Riddet, Craig; MacLaren, Donald A.; Asenov, Asen; Paul, Douglas J.

被引用次数: 20

2017

A Device-Level Characterization Approach to Quantify the Impacts of Different Random Variation Sources in FinFET Technology

IEEE ELECTRON DEVICE LETTERS

Jiang, Xiaobo; Guo, Shaofeng; Wang, Runsheng; Wang, Xingsheng; Cheng, Binjie; Asenov, Asen; Huang, Ru

被引用次数: 20

2016

Multivariate Modeling of Variability Supporting Non-Gaussian and Correlated Parameters

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Lange, Andre; Sohrmann, Christoph; Jancke, Roland; Haase, Joachim; Cheng, Binjie; Asenov, Asen; Schlichtmann, Ulf

被引用次数: 13

2016

Insight Into Electron Traps and Their Energy Distribution Under Positive Bias Temperature Stress and Hot Carrier Aging

IEEE TRANSACTIONS ON ELECTRON DEVICES

Duan, Meng; Zhang, Jian Fu; Ji, Zhigang; Zhang, Wei Dong; Vigar, David; Asenov, Asen; Gerrier, Louis; Chandra, Vikas; Aitken, Rob; Kaczer, Ben

被引用次数: 24

2016

ICMAT 2015

MICROELECTRONICS RELIABILITY

Asenov, Asen; Schichtmann, Ulf; Tan, Cher Ming; Wong, Hei; Zhou, Xing

被引用次数: 0

2016

Simulation Study of the Impact of Quantum Confinement on the Electrostatically Driven Performance of n-type Nanowire Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wang, Yijiao; Al-Ameri, Talib; Wang, Xingsheng; Georgiev, Vihar P.; Towie, Ewan; Amoroso, Salvatore Maria; Brown, Andrew R.; Cheng, Binjie; Reid, David; Riddet, Craig; Shifren, Lucian; Sinha, Saurabh; Yeric, Greg; Aitken, Robert; Liu, Xiaoyan; Kang, Jinfeng; Asenov, Asen

被引用次数: 22

2015

New Assessment Methodology Based on Energy-Delay-Yield Cooptimization for Nanoscale CMOS Technology

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jiang, Xiaobo; Wang, Junyao; Wang, Xingsheng; Wang, Runsheng; Cheng, Binjie; Asenov, Asen; Wei, Lan; Huang, Ru

被引用次数: 4

2015

Mastering CMOS variability is the key to success

IET COMPUTERS AND DIGITAL TECHNIQUES

Asenov, Asen

被引用次数: 0

2015

Influence of Transistors With BTI-Induced Aging on SRAM Write Performance

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ding, Jie; Reid, Dave; Asenov, Plamen; Millar, Campbell; Asenov, Asen

被引用次数: 5

2015

FinFET Centric Variability-Aware Compact Model Extraction and Generation Technology Supporting DTCO

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wang, Xingsheng; Cheng, Binjie; Reid, David; Pender, Andrew; Asenov, Plamen; Millar, Campbell; Asenov, Asen

被引用次数: 25

2015

Impact of Self-Heating on the Statistical Variability in Bulk and SOI FinFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wang, Liping; Brown, Andrew R.; Nedjalkov, Mihail; Alexander, Craig; Cheng, Binjie; Millar, Campbell; Asenov, Asen

被引用次数: 29

2015

Accurate Simulation of Transistor-Level Variability for the Purposes of TCAD-Based Device-Technology Cooptimization

IEEE TRANSACTIONS ON ELECTRON DEVICES

Gerrer, Louis; Brown, Andrew R.; Millar, Campbell; Hussin, Razaidi; Amoroso, Salvatore Maria; Cheng, Binjie; Reid, Dave; Alexander, Craig; Fried, David; Hargrove, Michael; Greiner, Ken; Asenov, Asen

被引用次数: 18

2015

A Mobility Correction Approach for Overcoming Artifacts in Atomistic Drift-Diffusion Simulation of Nano-MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Amoroso, Salvatore Maria; Adamu-Lema, Fikru; Brown, Andrew R.; Asenov, Asen

被引用次数: 4

2015

Variability Aware Simulation Based Design-Technology Cooptimization (DTCO) Flow in 14 nm FinFET/SRAM Cooptimization

IEEE TRANSACTIONS ON ELECTRON DEVICES

Asenov, Asen; Cheng, Binjie; Wang, Xingsheng; Brown, Andrew Robert; Millar, Campbell; Alexander, Craig; Amoroso, Salvatore Maria; Kuang, Jente B.; Nassif, Sani R.

被引用次数: 51

2015

Comparison Between Bulk and FDSOI POM Flash Cell: A Multiscale Simulation Study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Georgiev, Vihar P.; Amoroso, Salvatore Maria; Ali, Talib Mahmood; Vila-Nadal, Laia; Busche, Christoph; Cronin, Leroy; Asenov, Asen

被引用次数: 6

2015

Electron dynamics in nanoscale transistors by means of Wigner and Boltzmann approaches

PHYSICA A-STATISTICAL MECHANICS AND ITS APPLICATIONS

Sellier, J. M.; Amoroso, S. M.; Nedjalkov, M.; Selberherr, S.; Asenov, A.; Dimov, I.

被引用次数: 9

2014

Drain bias effects on statistical variability and reliability and related subthreshold variability in 20-nm bulk planar MOSFETs

SOLID-STATE ELECTRONICS

Wang, Xingsheng; Brown, Andrew R.; Cheng, Binjie; Roy, Scott; Asenov, Asen

被引用次数: 4

2014

Inverse Scaling Trends for Charge-Trapping-Induced Degradation of FinFETs Performance

IEEE TRANSACTIONS ON ELECTRON DEVICES

Amoroso, Salvatore Maria; Georgiev, Vihar P.; Gerrer, Louis; Towie, Ewan; Wang, Xingsheng; Riddet, Craig; Brown, Andrew Robert; Asenov, Asen

被引用次数: 7

2014

Performance and Variability of Doped Multithreshold FinFETs for 10-nm CMOS

IEEE TRANSACTIONS ON ELECTRON DEVICES

Adamu-Lema, Fikru; Wang, Xingsheng; Amoroso, Salvatore Maria; Riddet, Craig; Cheng, Binjie; Shifren, Lucian; Aitken, Robert; Sinha, Saurabh; Yeric, Greg; Asenov, Asen

被引用次数: 16

2014

Interplay Between Statistical Variability and Reliability in Contemporary pMOSFETs: Measurements Versus Simulations

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hussin, Razaidi; Amoroso, Salvatore Maria; Gerrer, Louis; Kaczer, Ben; Weckx, Pieter; Franco, Jacopo; Vanderheyden, Annelies; Vanhaeren, Danielle; Horiguchi, Naoto; Asenov, Asen

被引用次数: 8

2014

Development of a Technique for Characterizing Bias Temperature Instability-Induced Device-to-Device Variation at SRAM-Relevant Conditions

IEEE TRANSACTIONS ON ELECTRON DEVICES

Duan, Meng; Zhang, Jian Fu; Ji, Zhigang; Zhang, Wei Dong; Kaczer, Ben; Schram, Tom; Ritzenthaler, Romain; Groeseneken, Guido; Asenov, Asen

被引用次数: 21

2014

Problems With the Continuous Doping TCAD Simulations of Decananometer CMOS Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Asenov, Asen; Adamu-Lema, Fikru; Wang, Xingsheng; Amoroso, Salvatore Maria

被引用次数: 17

2014

Predictive Simulation and Benchmarking of Si and Ge pMOS FinFETs for Future CMOS Technology

IEEE TRANSACTIONS ON ELECTRON DEVICES

Shifren, Lucian; Aitken, Robert; Brown, Andrew R.; Chandra, Vikas; Cheng, Binjie; Riddet, Craig; Alexander, Craig L.; Cline, Brian; Millar, Campbell; Sinha, Saurabh; Yeric, Greg; Asenov, Asen

被引用次数: 21

2014

Time-Dependent 3-D Statistical KMC Simulation of Reliability in Nanoscale MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Amoroso, Salvatore Maria; Gerrer, Louis; Hussin, Razaidi; Adamu-Lema, Fikru; Asenov, Asen

被引用次数: 2

2014

Optimization and Evaluation of Variability in the Programming Window of a Flash Cell With Molecular Metal-Oxide Storage

IEEE TRANSACTIONS ON ELECTRON DEVICES

Georgiev, Vihar P.; Markov, Stanislav; Vila-Nadal, Laia; Busche, Christoph; Cronin, Leroy; Asenov, Asen

被引用次数: 15

2014

Modeling Carrier Mobility in Nano-MOSFETs in the Presence of Discrete Trapped Charges: Accuracy and Issues

IEEE TRANSACTIONS ON ELECTRON DEVICES

Amoroso, Salvatore Maria; Gerrer, Louis; Nedjalkov, Mihail; Hussin, Razaidi; Alexander, Craig; Asenov, Asen

被引用次数: 3

2014

RTN distribution comparison for bulk, FDSOI and FinFETs devices

MICROELECTRONICS RELIABILITY

Gerrer, L.; Amoroso, S. M.; Hussin, R.; Asenov, A.

被引用次数: 11

2014

Special section reliability and variability of devices for circuits and systems

MICROELECTRONICS RELIABILITY

Asenov, Asen; Schlichtmann, Ulf; Tan, Cher Ming; Wong, Hei; Zhou, Xing

被引用次数: 0

2014

Modelling RTN and BTI in nanoscale MOSFETs from device to circuit: A review

MICROELECTRONICS RELIABILITY

Gerrer, L.; Ding, J.; Amoroso, S. M.; Adamu-Lema, F.; Hussin, R.; Reid, D.; Millar, C.; Asenov, A.

被引用次数: 31

2014

Design and fabrication of memory devices based on nanoscale polyoxometalate clusters

NATURE

Busche, Christoph; Vila-Nadal, Laia; Yan, Jun; Miras, Haralampos N.; Long, De-Liang; Georgiev, Vihar P.; Asenov, Asen; Pedersen, Rasmus H.; Gadegaard, Nikolaj; Mirza, Muhammad M.; Paul, Douglas J.; Poblet, Josep M.; Cronin, Leroy

被引用次数: 292

2014

Back-Gate Bias Dependence of the Statistical Variability of FDSOI MOSFETs With Thin BOX

IEEE TRANSACTIONS ON ELECTRON DEVICES

Yang, Yunxiang; Markov, Stanislav; Cheng, Binjie; Zain, Anis Suhaila Mohd; Liu, Xiaoyan; Asenov, Asen

被引用次数: 16

2013

Comprehensive study of the statistical variability in a 22 nm fully depleted ultra-thin-body SOI MOSFET

SOLID-STATE ELECTRONICS

Zain, Anis Suhaila Mohd; Markov, Stanislav; Cheng, Binjie; Asenov, Asen

被引用次数: 9

2013

RTN and BTI in nanoscale MOSFETs: A comprehensive statistical simulation study

SOLID-STATE ELECTRONICS

Amoroso, Salvatore Maria; Gerrer, Louis; Markov, Stanislav; Adamu-Lema, Fikru; Asenov, Asen

被引用次数: 17

2013

Analytical Models for Three-Dimensional Ion Implantation Profiles in FinFETs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Wang, Liping; Brown, Andrew R.; Cheng, Binjie; Asenov, Asen

被引用次数: 6

2013

Towards Polyoxometalate-Cluster-Based Nano-Electronics

CHEMISTRY-A EUROPEAN JOURNAL

Vila-Nadal, Laia; Mitchell, Scott G.; Markov, Stanislav; Busche, Christoph; Georgiev, Vihar; Asenov, Asen; Cronin, Leroy

被引用次数: 66

2013

Investigation of the RTN Distribution of Nanoscale MOS Devices From Subthreshold to On-State

IEEE ELECTRON DEVICE LETTERS

Amoroso, Salvatore M.; Compagnoni, Christian Monzio; Ghetti, Andrea; Gerrer, Louis; Spinelli, Alessandro S.; Lacaita, Andrea L.; Asenov, Asen

被引用次数: 35

2013

Statistical Interactions of Multiple Oxide Traps Under BTI Stress of Nanoscale MOSFETs

IEEE ELECTRON DEVICE LETTERS

Markov, Stanislav; Amoroso, Salvatore Maria; Gerrer, Louis; Adamu-Lema, Fikru; Asenov, Asen

被引用次数: 15

2013

3-D Statistical Simulation Comparison of Oxide Reliability of Planar MOSFETs and FinFET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Gerrer, Louis; Amoroso, Salvatore Maria; Markov, Stanislav; Adamu-Lema, Fikru; Asenov, Asen

被引用次数: 22

2013

Comparative Simulation Analysis of Process-Induced Variability in Nanoscale SOI and Bulk Trigate FinFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Brown, Andrew R.; Daval, Nicolas; Bourdelle, Konstantin K.; Bich-Yen Nguyen; Asenov, Asen

被引用次数: 37

2013

Interplay Between Process-Induced and Statistical Variability in 14-nm CMOS Technology Double-Gate SOI FinFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wang, Xingsheng; Cheng, Binjie; Brown, Andrew Robert; Millar, Campbell; Kuang, Jente B.; Nassif, Sani; Asenov, Asen

New Analysis Method for Time-Dependent Device-To-Device Variation Accounting for Within-Device Fluctuation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Duan, Meng; Zhang, Jian F.; Ji, Zhigang; Zhang, Wei Dong; Kaczer, Ben; Schram, Tom; Ritzenthaler, Romain; Groeseneken, Guido; Asenov, Asen

被引用次数: 31

2013

Geometry, Temperature, and Body Bias Dependence of Statistical Variability in 20-nm Bulk CMOS Technology: A Comprehensive Simulation Analysis

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wang, Xingsheng; Adamu-Lema, Fikru; Cheng, Binjie; Asenov, Asen

被引用次数: 29

2013

Impact of Precisely Positioned Dopants on the Performance of an Ultimate Silicon Nanowire Transistor: A Full Three-Dimensional NEGF Simulation Study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Georgiev, Vihar P.; Towie, Ewan A.; Asenov, Asen

被引用次数: 24

2013

Accuracy and Issues of the Spectroscopic Analysis of RTN Traps in Nanoscale MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Adamu-Lema, Fikru; Compagnoni, Christian Monzio; Amoroso, Salvatore M.; Castellani, Niccolo; Gerrer, Louis; Markov, Stanislav; Spinelli, Alessandro S.; Lacaita, Andrea L.; Asenov, Asen

被引用次数: 25

2013

Impact of statistical parameter set selection on the statistical compact model accuracy: BSIM4 and PSP case study

MICROELECTRONICS JOURNAL

Moezi, Negin; Dideban, Daryoosh; Cheng, Binjie; Roy, Scott; Asenov, Asen

被引用次数: 8

2013

Statistical Variability and Reliability and the Impact on Corresponding 6T-SRAM Cell Design for a 14-nm Node SOI FinFET Technology

IEEE DESIGN & TEST

Wang, Xingsheng; Cheng, Binjie; Brown, Andrew R.; Millar, Campbell; Kuang, Jente B.; Nassif, Sani; Asenov, Asen

被引用次数: 17

2013

From atoms to product reliability: toward a generalized multiscale simulation approach

JOURNAL OF COMPUTATIONAL ELECTRONICS

Gerrer, Louis; Ling, Sanliang; Amoroso, Salvatore Maria; Asenov, Plamen; Shluger, Alexandre L.; Asenov, Asen

被引用次数: 8

2013

NEGF simulations of a junctionless Si gate-all-around nanowire transistor with discrete dopants

SOLID-STATE ELECTRONICS

Martinez, A.; Aldegunde, M.; Brown, A. R.; Roy, S.; Asenov, A.

被引用次数: 23

2012

Statistical Variability in Fully Depleted SOI MOSFETs Due to Random Dopant Fluctuations in the Source and Drain Extensions

IEEE ELECTRON DEVICE LETTERS

Markov, Stanislav; Cheng, Binjie; Asenov, Asen

被引用次数: 47

2012

Enriched residual free bubbles for semiconductor device simulation

COMPUTATIONAL MECHANICS

Simpson, R. N.; Bordas, S. P. A.; Asenov, A.; Brown, A. R.

被引用次数: 9

2012

Impact of Cell Shape on Random Telegraph Noise in Decananometer Flash Memories

IEEE TRANSACTIONS ON ELECTRON DEVICES

Amoroso, Salvatore Maria; Ghetti, Andrea; Brown, Andrew R.; Mauri, Aurelio; Compagnoni, Christian Monzio; Asenov, Asen

被引用次数: 9

2012

Hole Mobility in Germanium as a Function of Substrate and Channel Orientation, Strain, Doping, and Temperature

IEEE TRANSACTIONS ON ELECTRON DEVICES

Riddet, Craig; Watling, Jeremy R.; Chan, Kah-Hou; Parker, Evan H. C.; Whall, Terence E.; Leadley, David R.; Asenov, Asen

被引用次数: 28

2012

Accurate capturing of the statistical aspect of NBTI/PBTI variability into statistical compact models

MICROELECTRONICS JOURNAL

Bukhori, Muhammad Faiz; Kamsani, Noor Ain; Asenov, Asen; Nayan, Nazrul Anuar

被引用次数: 1

2012

ICMAT 2011-Reliability and variability of semiconductor devices and ICs

MICROELECTRONICS RELIABILITY

Asenov, Asen; Schlichtmann, Ulf; Tan, Cher Ming; Wong, Hei; Zhou, Xing

被引用次数: 0

2012

FET based nano-pore sensing: a 3D simulation study

JOURNAL OF COMPUTATIONAL ELECTRONICS

Moore, I.; Millar, C.; Roy, S.; Asenov, A.

被引用次数: 2

2012

Numerical analysis of the new Implant-Free Quantum-Well CMOS: DualLogic approach

SOLID-STATE ELECTRONICS

Benbakhti, Brahim; Chan, KahHou; Towie, Ewan; Kalna, Karol; Riddet, Craig; Wang, Xingsheng; Eneman, Geert; Hellings, Geert; De Meyer, Kristin; Meuris, Marc; Asenov, Asen

被引用次数: 3

2011

Implementation of the Density Gradient Quantum Corrections for 3-D Simulations of Multigate Nanoscaled Transistors

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Garcia-Loureiro, Antonio J.; Seoane, Natalia; Aldegunde, Manuel; Valin, Raul; Asenov, Asen; Martinez, Antonio; Kalna, Karol

被引用次数: 58

2011

The evolution of standard cell libraries for future technology nodes

GENETIC PROGRAMMING AND EVOLVABLE MACHINES

Walker, James Alfred; Hilder, James A.; Reid, Dave; Asenov, Asen; Roy, Scott; Millar, Campbell; Tyrrell, Andy M.

被引用次数: 9

2011

Impact of NBTI/PBTI on SRAM Stability Degradation

IEEE ELECTRON DEVICE LETTERS

Cheng, Binjie; Brown, Andrew R.; Asenov, Asen

被引用次数: 41

2011

Impact of STI on Statistical Variability and Reliability of Decananometer MOSFETs

IEEE ELECTRON DEVICE LETTERS

Wang, Xingsheng; Roy, Scott; Brown, Andrew R.; Asenov, Asen

被引用次数: 14

2011

Comparative Simulation Study of the Different Sources of Statistical Variability in Contemporary Floating-Gate Nonvolatile Memory

IEEE TRANSACTIONS ON ELECTRON DEVICES

Roy, Gareth; Ghetti, Andrea; Benvenuti, Augusto; Erlebach, Axel; Asenov, Asen

被引用次数: 19

2011

Foreword Special Issue on Characterization of Nano CMOS Variability by Simulation and Measurements

IEEE TRANSACTIONS ON ELECTRON DEVICES

Sangiorgi, Enrico; Asenov, Asen; Bennett, Herbert S.; Dutton, Robert W.; Esseni, David; Giles, Martin D.; Hane, Masami; Nishi, Kenji; Ranaweera, Jeewika; Selberherr, Siegfried

被引用次数: 1

2011

Quantum-Transport Study on the Impact of Channel Length and Cross Sections on Variability Induced by Random Discrete Dopants in Narrow Gate-All-Around Silicon Nanowire Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Martinez, Antonio; Aldegunde, Manuel; Seoane, Natalia; Brown, Andrew R.; Barker, John R.; Asenov, Asen
被引用次数: 72 2011

Statistical Threshold-Voltage Variability in Scaled Decananometer Bulk HKMG MOSFETs: A Full-Scale 3-D Simulation Scaling Study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wang, Xingsheng; Brown, Andrew R.; Idris, Niza; Markov, Stanislav; Roy, Gareth; Asenov, Asen

被引用次数: 122 2011

Drain Current Collapse in Nanoscaled Bulk MOSFETs Due to Random Dopant Compensation in the Source/Drain Extensions

IEEE TRANSACTIONS ON ELECTRON DEVICES

Markov, Stanislav; Wang, Xingsheng; Moezi, Negin; Asenov, Asen

被引用次数: 7 2011

Simulation study of the 20 nm gate-length Ge implant-free quantum well p-MOSFET

MICROELECTRONIC ENGINEERING

Chan, K. H.; Benbakhti, B.; Riddet, C.; Watling, J. R.; Asenov, A.

被引用次数: 0 2011

Simulation of hole-mobility in doped relaxed and strained Ge

MICROELECTRONIC ENGINEERING

Watling, Jeremy R.; Riddet, Craig; Chan, Kah H.; Asenov, Asen

被引用次数: 1 2011

Simulation of hole-mobility in doped relaxed and strained Ge layers

JOURNAL OF APPLIED PHYSICS

Watling, Jeremy R.; Riddet, Craig; Chan, Morgan Kah H.; Asenov, Asen

被引用次数: 4 2010

Parameter set and data sampling strategy for accurate yet efficient statistical MOSFET compact model extraction

SOLID-STATE ELECTRONICS

Bindu, B.; Cheng, B.; Roy, G.; Wang, X.; Roy, S.; Asenov, A.

被引用次数: 8 2010

Impact of Metal Gate Granularity on Threshold Voltage Variability: A Full-Scale Three-Dimensional Statistical Simulation Study

IEEE ELECTRON DEVICE LETTERS

Brown, Andrew R.; Idris, Niza M.; Watling, Jeremy R.; Asenov, Asen

被引用次数: 101 2010

PBTI/NBTI-Related Variability in TB-SOI and DG MOSFETs

IEEE ELECTRON DEVICE LETTERS

Cheng, B.; Brown, A. R.; Roy, S.; Asenov, A.

被引用次数: 15 2010

Direct Tunnelling Gate Leakage Variability in Nano-CMOS Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Markov, Stanislav; Roy, Scott; Asenov, Asen

被引用次数: 15

2010

Statistical Simulation of Progressive NBTI Degradation in a 45-nm Technology pMOSFET

IEEE TRANSACTIONS ON ELECTRON DEVICES

Brown, Andrew R.; Huard, Vincent; Asenov, Asen

被引用次数: 25

2010

Variability in Si Nanowire MOSFETs Due to the Combined Effect of Interface Roughness and Random Dopants: A Fully Three-Dimensional NEGF Simulation Study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Martinez, Antonio; Seoane, Natalia; Brown, Andrew R.; Barker, John R.; Asenov, Asen

被引用次数: 45

2010

Simulation of Statistical Aspects of Charge Trapping and Related Degradation in Bulk MOSFETs in the Presence of Random Discrete Dopants

IEEE TRANSACTIONS ON ELECTRON DEVICES

Bukhori, Muhammad Faiz; Roy, Scott; Asenov, Asen

被引用次数: 38

2010

Use of density gradient quantum corrections in the simulation of statistical variability in MOSFETs

JOURNAL OF COMPUTATIONAL ELECTRONICS

Brown, Andrew R.; Watling, Jeremy R.; Roy, Gareth; Riddet, Craig; Alexander, Craig L.; Kovac, Urban; Martinez, Antonio; Asenov, Asen

被引用次数: 29

2010

A comparison of advanced transport models for the computation of the drain current in nanoscale nMOSFETs

SOLID-STATE ELECTRONICS

Palestri, P.; Alexander, C.; Asenov, A.; Aubry-Fortuna, V.; Baccarani, G.; Bournel, A.; Braccioli, M.; Cheng, B.; Dollfus, P.; Esposito, A.; Esseni, D.; Fenouillet-Beranger, C.; Fiegna, C.; Fiori, G.; Ghetti, A.; Iannaccone, G.; Martinez, A.; Majkusiak, B.; Monfray, S.; Peikert, V.; Reggiani, S.; Riddet, C.; Saint-Martin, J.; Sangiorgi, E.; Schenk, A.; Selmi, L.; Silvestri, L.; Toniutti, P.; Walczak, J.

被引用次数: 13

2009

Evaluation of statistical variability in 32 and 22 nm technology generation LSTP MOSFETs

SOLID-STATE ELECTRONICS

Cheng, B.; Roy, S.; Brown, A. R.; Millar, C.; Asenov, A.

被引用次数: 23

2009

Analysis of Threshold Voltage Distribution Due to Random Dopants: A 100 000-Sample 3-D Simulation Study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Reid, Dave; Millar, Campbell; Roy, Gareth; Roy, Scott; Asenov, Asen

被引用次数: 58

2009

Current Variability in Si Nanowire MOSFETs Due to Random Dopants in the Source/Drain Regions: A Fully 3-D NEGF Simulation Study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Seoane, Natalia; Martinez, Antonio; Brown, Andrew R.; Barker, John R.; Asenov, Asen

被引用次数: 67

2009

3-D Nonequilibrium Green's Function Simulation of Nonperturbative Scattering From Discrete Dopants in the Source and Drain of a Silicon Nanowire Transistor

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Martinez, Antonio; Seoane, Natalia; Brown, Andrew R.; Barker, John R.; Asenov, Asen

被引用次数: 27

2009

Impact of Body-Thickness-Dependent Band Structure on Scaling of Double-Gate MOSFETs: A DFT/NEGF Study

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Martinez, Antonio; Kalna, Karol; Sushko, Peter V.; Shluger, Alex L.; Barker, John R.; Asenov, Asen

被引用次数: 24

2009

Enabling cutting-edge semiconductor simulation through grid technology

PHILOSOPHICAL TRANSACTIONS OF THE ROYAL SOCIETY A-MATHEMATICAL PHYSICAL AND ENGINEERING SCIENCES

Reid, Dave; Millar, Campbell; Roy, Scott; Roy, Gareth; Sinnott, Richard; Stewart, Gordon; Stewart, Graeme; Asenov, Asen

被引用次数: 3

2009

Atomistic mesh generation for the simulation of nanoscale metal-oxide-semiconductor field-effect transistors

PHYSICAL REVIEW E

Aldegunde, M.; Seoane, Natalia; Garcia-Loureiro, A. J.; Sushko, P. V.; Shluger, A. L.; Gavartin, J. L.; Kalna, K.; Asenov, A.

被引用次数: 3

2008

Accurate statistical description of random dopant-induced threshold voltage variability

IEEE ELECTRON DEVICE LETTERS

Millar, Campbell; Reid, David; Roy, Gareth; Roy, Scott; Asenov, Asen

被引用次数: 51

2008

Benchmarking of scaled InGaAs implant-free NanoMOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kalna, Karol; Seoane, Natalia; Garcia-Loureiro, Antonio J.; Thayne, Iain G.; Asenov, Asen

被引用次数: 39

2008

Statistical aspects of reliability in bulk MOSFETs with multiple defect states and random discrete dopants

MICROELECTRONICS RELIABILITY

Bukhori, Muhammad Faiz; Roy, Scott; Asenov, Asen

被引用次数: 19

2008

Statistical simulation of random dopant induced threshold voltage fluctuations for 35 nm channel length MOSFET

MICROELECTRONICS RELIABILITY

Kovac, Urban; Reid, Dave; Millar, Campbell; Roy, Gareth; Roy, Scott; Asenov, Asen

被引用次数: 10

2008

Capacitance fluctuations in bulk MOSFETs due to random discrete dopants

JOURNAL OF COMPUTATIONAL ELECTRONICS

Brown, Andrew R.; Asenov, Asen

被引用次数: 17

2008

Capacitance variability of short range interconnects

JOURNAL OF COMPUTATIONAL ELECTRONICS

Drysdale, Timothy D.; Brown, Andrew R.; Roy, Gareth; Roy, Scott; Asenov, Asen

被引用次数: 7

2008

Boundary conditions for Density Gradient corrections in 3D Monte Carlo simulations

JOURNAL OF COMPUTATIONAL ELECTRONICS

Riddet, Craig; Brown, Andrew R.; Roy, Scott; Asenov, Asen

被引用次数: 9

2008

A full 3D non-equilibrium Green functions study of a stray charge in a nanowire MOS transistor

JOURNAL OF COMPUTATIONAL ELECTRONICS

Martinez, A.; Bescond, M.; Brown, A. R.; Barker, J. R.; Asenov, A.

被引用次数: 3

2008

Brownian simulation of charge transport in α -Haemolysin

JOURNAL OF COMPUTATIONAL ELECTRONICS

Millar, C.; Madathil, R.; Beckstein, O.; Sansom, M. S. P.; Roy, S.; Asenov, A.

被引用次数: 4

2008

Investigation into effects of device variability on CMOS layout motifs

ELECTRONICS LETTERS

Paluchowski, S. H.; Cheng, B.; Roy, S.; Asenov, A.; Cumming, D. R. S.

被引用次数: 4

2008

A study of the interface roughness effect in Si nanowires using a full 3D NEGF approach

PHYSICA E-LOW-DIMENSIONAL SYSTEMS & NANOSTRUCTURES

Martinez, A.; Kalna, K.; Barker, J. R.; Asenov, A.

被引用次数: 9

2007

Combined sources of intrinsic parameter fluctuations in sub-25 nm generation UTB-SOI MOSFETs: A statistical simulation study

SOLID-STATE ELECTRONICS

Samsudin, K.; Adamu-Lerna, F.; Brown, A. R.; Roy, S.; Asenov, A.

被引用次数: 18

2007

Impact of intrinsic parameter fluctuations on the performance of HEMTs studied with a 3D parallel drift-diffusion simulator

SOLID-STATE ELECTRONICS

Seoane, N.; Garcia-Loureiro, A. J.; Kalna, K.; Asenov, A.

被引用次数: 9

2007

Poly-Si-gate-related variability in decananometer MOSFETs with conventional architecture

IEEE TRANSACTIONS ON ELECTRON DEVICES

Brown, Andrew R.; Roy, Gareth; Asenov, Asen

被引用次数: 84

2007

A self-consistent full 3-D real-space NEGF simulator for studying nonperturbative effects in nano-MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Martinez, Antonio; Bescond, Marc; Barker, John R.; Svizhenko, Alexei; Anantram, M. P.; Millar, Campbell; Asenov, Asen

被引用次数: 129

2007

The impact of random dopant aggregation in source and drain on the performance of ballistic DG Nano-MOSFETs: A NEGF study

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Martinez, Antonio; Barker, John R.; Svizhenko, Alexei; Anantram, M. P.; Asenov, Asen

被引用次数: 27

2007

3-D Monte Carlo simulation of the impact of quantum confinement scattering on the magnitude of current fluctuations in double gate MOSFETs

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Riddet, Craig; Brown, Andrew R.; Alexander, Craig L.; Watling, Jeremy R.; Roy, Scott; Asenov, Asen

被引用次数: 29

2007

Monte Carlo simulations of InGaAs nano-MOSFETs

MICROELECTRONIC ENGINEERING

Kalna, K.; Droopad, R.; Passlack, M.; Asenov, A.

被引用次数: 8

2007

Simulation of implant free III-V MOSFETs for high performance low power Nano-CMOS applications

MICROELECTRONIC ENGINEERING

Asenov, A.; Kalna, K.; Thayne, I.; Hill, R. J. W.

On the impact of high- κ gate stacks on mobility: A Monte Carlo study including coupled SO phonon-plasmon scattering

JOURNAL OF COMPUTATIONAL ELECTRONICS

Ferrari, Giulio; Watling, J. R.; Roy, S.; Barker, J. R.; Asenov, A.; Zeitzoff, P.; Bersuker, G.

被引用次数: 4

2007

Developing a full 3D NEGF simulator with random dopant and interface roughness

JOURNAL OF COMPUTATIONAL ELECTRONICS

Martinez, A.; Barker, J. R.; Asenov, A.; Svizhenko, A.; Anantram, M. P.

被引用次数: 9

2007

Continuum vs. particle simulations of model nano-pores

JOURNAL OF COMPUTATIONAL ELECTRONICS

Millar, C.; Roy, S.; Beckstein, O.; Sansom, M. S. P.; Asenov, A.

被引用次数: 1

2007

Integrating intrinsic parameter fluctuation description into BSIMSOI to forecast sub-15 nm UTB SOI based 6T SRAM operation

SOLID-STATE ELECTRONICS

Samsudin, K; Cheng, B; Brown, AR; Roy, S; Asenov, A

被引用次数: 63

2006

Simulation study of individual and combined sources of intrinsic parameter fluctuations in conventional nano-MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Roy, Gareth; Brown, Andrew R.; Adamu-Lema, Fikru; Roy, Scott; Asenov, Asen

被引用次数: 235

2006

Monte Carlo study of mobility in Si devices with HfO₂-based oxides

MATERIALS SCIENCE IN SEMICONDUCTOR PROCESSING

Ferrari, Giulio; Watling, J. R.; Roy, S.; Barker, J. R.; Zeitzoff, P.; Bersuker, G.; Asenov, A.

被引用次数: 5

2006

Study of fluctuations in advanced MOSFETs using a 3D finite element parallel simulator

JOURNAL OF COMPUTATIONAL ELECTRONICS

Aldegunde, M.; Garcia-Loureiro, A. J.; Kalna, K.; Asenov, A.

被引用次数: 4

2006

Intrinsic parameter fluctuations due to random grain orientations in high- κ gate stacks

JOURNAL OF COMPUTATIONAL ELECTRONICS

Brown, Andrew R.; Watling, Jeremy R.; Asenov, Asen

被引用次数: 25

2006

Statistical study of the effect of interface charge fluctuations in HEMTs using a 3D simulator

JOURNAL OF COMPUTATIONAL ELECTRONICS

Seoane, N.; Garcia-Loureiro, A. J.; Kalna, K.; Asenov, A.

被引用次数: 3

2006

Introducing energy broadening in semiclassical Monte Carlo simulations

JOURNAL OF COMPUTATIONAL ELECTRONICS

Ferrari, Giulio; Asenov, A.; Nedjalkov, M.; Jacoboni, C.

被引用次数: 2

2006

Atomistic effect of delta doping layer in a 50 nm InP HEMT

JOURNAL OF COMPUTATIONAL ELECTRONICS

Seoane, N.; Garcia-Loureiro, A. J.; Kalna, K.; Asenov, A.

被引用次数: 3

2006

Impact of intrinsic parameter fluctuations in decanano MOSFETs on yield and functionality of SRAM cells

SOLID-STATE ELECTRONICS

Cheng, B; Roy, S; Roy, G; Adamu-Lema, F; Asenov, A

被引用次数: 70

2005

Where do the dopants go?

SCIENCE

Roy, S; Asenov, A

被引用次数: 114

2005

The impact of interface roughness scattering and degeneracy in relaxed and strained Si n-channel MOSFETs

SOLID-STATE ELECTRONICS

Watling, JR; Yang, L; Boriçi, M; Wilkins, RCW; Asenov, A; Barker, JR; Roy, S

被引用次数: 41

2004

Role of multiple delta doping in PHEMTs scaled to sub-100 nm dimensions

SOLID-STATE ELECTRONICS

Kalna, K; Asenov, A

被引用次数: 18

2004

Impact of device geometry and doping strategy on linearity and RF performance in Si/SiGe MODFETs

MICROELECTRONICS RELIABILITY

Yang, L; Asenov, A; Watling, JR; Boriçi, M; Barker, JR; Roy, S; Elgaid, K; Thayne, I; Hackbarth, T

被引用次数: 1

2004

Simulations of Sub-100 nm Strained Si MOSFETs with High- κ Gate Stacks

JOURNAL OF COMPUTATIONAL ELECTRONICS

Yang, L; Watling, J. R.; Adamu-Lema, F.; Asenov, A.; Barker, J. R.

被引用次数: 1

2004

Intrinsic Parameter Fluctuations in Conventional MOSFETs at the Scaling Limit: A Statistical Study

JOURNAL OF COMPUTATIONAL ELECTRONICS

Adamu-Lema, Fikru; Roy, Gareth; Brown, Andrew R.; Asenov, Asen; Roy, Scott

被引用次数: 1

2004

Random Telegraph Noise in 30 nm FETs with Conventional and High- κ Dielectrics

JOURNAL OF COMPUTATIONAL ELECTRONICS

Lee, Angelica; Brown, Andrew R.; Asenov, Asen; Roy, Scott

被引用次数: 3

2004

Si/SiGe heterostructure parameters for device simulations

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Yang, LF; Watling, JR; Wilkins, RCW; Boriçi, M; Barker, JR; Asenov, A; Roy, S

被引用次数: 155

2004

Numerical carrier heating when implementing P3M to study small volume variations

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Alexander, C; Watling, J; Asenov, A

被引用次数: 4

2004

Interface roughness scattering and its impact on electron transport in relaxed and strained Si n-MOSFETs

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Boriçi, M; Watling, JR; Wilkins, RCW; Yang, L; Barker, JR; Asenov, A

被引用次数: 4

2004

Quantum corrections in the simulation of decanano MOSFETs

SOLID-STATE ELECTRONICS

Asenov, A; Brown, AR; Watling, JR

被引用次数: 43

2003

Nonequilibrium and ballistic transport, and backscattering in decanano HEMTs: a Monte Carlo simulation study

MATHEMATICS AND COMPUTERS IN SIMULATION

Kalna, K; Asenov, A

被引用次数: 2

2003

Simulation of intrinsic parameter fluctuations in decananometer and nanometer-scale MOSFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Asenov, A; Brown, AR; Davies, JH; Kaya, S; Slavcheva, G

被引用次数: 472

2003

Intrinsic parameter fluctuations in decananometer MOSFETs introduced by gate line edge roughness

IEEE TRANSACTIONS ON ELECTRON DEVICES

Asenov, A; Kaya, S; Brown, AR

被引用次数: 506

2003

RTS amplitudes in decananometer MOSFETs: 3-D Simulation Study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Asenov, A; Balasubramaniam, R; Brown, AR; Davies, JH

被引用次数: 221

2003

Brownian Ionic Channel Simulation

JOURNAL OF COMPUTATIONAL ELECTRONICS

Millar, C.; Asenov, A.; Roy, S.

被引用次数: 4

2003

Mobility Variations in Ultra Small Devices due to Random Discrete Dopants

JOURNAL OF COMPUTATIONAL ELECTRONICS

Alexander, Craig; Watling, Jeremy R.; Asenov, Asen

被引用次数: 0

2003

Simulation Study of High Performance III-V MOSFETs for Digital Applications

JOURNAL OF COMPUTATIONAL ELECTRONICS

Kalna, K.; Yang, L.; Asenov, A.

被引用次数: 3

2003

Study of RF Linearity in sub-50 nm MOSFETs Using Simulations

JOURNAL OF COMPUTATIONAL ELECTRONICS

Ma, Wei; Kaya, Savas; Asenov, Asen

被引用次数: 8

2003

Degeneracy and High Doping Effects in Deep Sub-Micron Relaxed and Strained Si n-MOSFETs

JOURNAL OF COMPUTATIONAL ELECTRONICS

Watling, J. R.; Yang, L.; Borici, M.; Barker, J. R.; Asenov, A.

被引用次数: 9

2003

Gate tunnelling and impact ionisation in sub 100 nm PHEMTs

IEICE TRANSACTIONS ON ELECTRONICS

Kalna, K; Asenov, A

被引用次数: 4

2003

Band offsets in III-nitride heterostructures

JOURNAL OF PHYSICS D-APPLIED PHYSICS

Ünlü, H; Asenov, A

被引用次数: 25

2002

Scaling of pseudomorphic high electron mobility transistors to decanano dimensions

SOLID-STATE ELECTRONICS

Kalna, K; Roy, S; Asenov, A; Elgaid, K; Thayne, I

被引用次数: 49

2002

Intrinsic threshold voltage fluctuations in decanano MOSFETs due to local oxide thickness variations

IEEE TRANSACTIONS ON ELECTRON DEVICES

Asenov, A; Kaya, S; Davies, JH

被引用次数: 177

2002

Intrinsic fluctuations in sub 10-nm double-gate MOSFETs introduced by discreteness of charge and matter

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Brown, AR; Asenov, A; Watling, JR

被引用次数: 88

2002

Breakdown of universal mobility curves in sub-100-nm MOSFETs

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Kaya, S; Asenov, A; Roy, S

被引用次数: 2

2002

Nonequilibrium transport in scaled high electron mobility transistors

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Kalna, K; Asenov, A

被引用次数: 9

2002

Effective mobilities in pseudomorphic Si/SiGe/Si p-channel metal-oxide-semiconductor field-effect transistors with thin silicon capping layers

APPLIED PHYSICS LETTERS

Palmer, MJ; Braithwaite, G; Grasby, TJ; Phillips, PJ; Prest, MJ; Parker, EHC; Whall, TE; Parry, CP; Waite, AM; Evans, AGR; Roy, S; Watling, JR; Kaya, S; Asenov, A

被引用次数: 47

2001

An electron emission model for use with 3D electromagnetic finite element simulation

SOLID-STATE ELECTRONICS

Knox, AR; Asenov, A; Lowe, AC

被引用次数: 6

2001

On the mobility extraction for HMOSFETs

SOLID-STATE ELECTRONICS

Straube, UN; Evans, AGR; Braithwaite, G; Kaya, S; Watling, J; Asenov, A

被引用次数: 1

2001

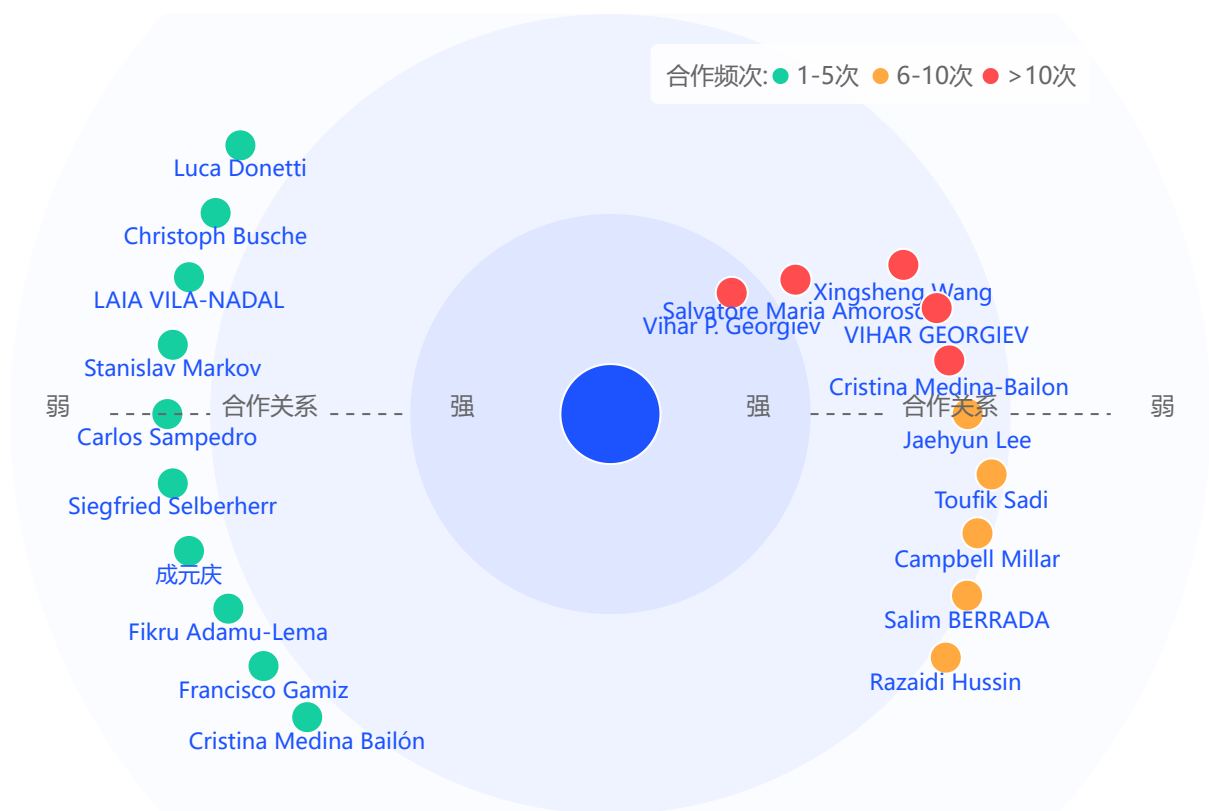
Increase in the random dopant induced threshold fluctuations and lowering in sub-100 nm MOSFETs due to quantum effects: A 3-D density-gradient simulation study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Asenov, A; Slavcheva, G; Brown, AR; Davies, JH; Saini, S

被引用次数: 177	2001
Strain engineered pHEMTs on virtual substrates: a Monte Carlo simulation study	
SOLID-STATE ELECTRONICS	
Babiker, S; Asenov, A; Roy, S; Beaumont, SP	
被引用次数: 8	1999
Optimization of layer structure for InGaAs channel pseudomorphic HEMTs	
JOURNAL OF CRYSTAL GROWTH	
Pearson, JL; Holland, MC; Stanley, CR; Long, AR; Skuras, E; Asenov, A; Davies, JH	
被引用次数: 7	1999
On the design and control of quantum effects in mesoscopic devices	
MICROELECTRONIC ENGINEERING	
Barker, JR; Asenov, A	
被引用次数: 0	1999
Mesh-based particle simulation of sub-0.1 micron FETs	
SEMICONDUCTOR SCIENCE AND TECHNOLOGY	
Arokianathan, C; Asenov, A; Davies, J	
被引用次数: 4	1998
HOT-CARRIER DEGRADATION MONITORING IN LDD N-MOSFETS USING DRAIN GATED-DIODE MEASUREMENTS	
MICROELECTRONIC ENGINEERING	
ASENOV, A; BERGER, J; WEBER, W; BOLLU, M; KOCH, F	
被引用次数: 8	1991
Multivariate Modeling of Variability Supporting Non-Gaussian and Correlated Parameters	
TCAD	
André Lange; Christoph Sohrmann; Roland Jancke; Joachim Haase; Binjie Cheng; Asen Asenov; Ulf Schlichtmann	
被引用次数: 0	

学术合作



Vihar P. Georgiev

格拉斯哥大学

合作次数: 23

Salvatore Maria Amoroso

格拉斯哥大学

合作次数: 19

Xingsheng Wang

格拉斯哥大学

合作次数: 12

VIHAR GEORGIEV

格拉斯哥大学

合作次数: 11

Cristina Medina-Bailon

格拉斯哥大学

合作次数: 11

还有 45 个合作者